19. Monday, May 12 Algebra of Sets (5.3), Cartesian Product (5.4)

Last Time: · Power Set $\mathcal{P}(A) = \{\text{subsets of } A\}$

· Element-Chasing To prove $A \subseteq B$

Theorem 1: Let A, B, C be subsets of a universal set U. Then

1) $A \cap B \subseteq A$ and $A \subseteq A \cup B$

2) If $A \subseteq B$ Then $A \cap C \subseteq B \cap C$ and $A \cup C \subseteq B \cup C$

Proof:

1) Suppose A and B are subsets of U. First we show $A \cap B \subseteq A$

Let $x \in A \cap B$. Then, by definition, $x \in A$ and $x \in B$. In particular, $x \in A$. Hence $A \cap B \subseteq A$

Now we show $A \subseteq A \cup B$.

Let $x \in A$. Then it is true that $x \in A$ or $x \in B$. Thus $x \in A \cup B$. This shows $A \subseteq A \cup B$.

2) Exercise! □

Theorem ("Algebra of sets Theorem") AST

Let A, B, C be subsets of a universal set U. Then the following hold:

1) $A \cap \phi = \phi$ and $A \cup \phi = A$

2) $A \cap U = A$ and $A \cup U = U$

3) Idempotent laws: $A \cap A = A$ and $A \cup A = A$

4) Commutative laws: $A \cap B = B \cap A$ and $A \cup B = B \cup A$

5) Associative laws: $(A \cap B) \cap C = A \cap (B \cap C)$ and $(A \cup B) \cup C = A \cup (B \cup C)$

6) Distributive laws: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ and $A \cup (B \cap C) = (A \cup B) \cap (A \cap C)$

7) $(A^c)^c = A$, $A \setminus A = \phi$, and $A \setminus B = A \cap B^c$

8) $A \setminus \phi = A$ and $A \setminus U = \phi$

9) $\phi^c = U$ and $U^c = \phi$

10) De Morgan's laws: $(A \cap B)^c = A^c \cup B^c$ and $(A \cup B)^c = A^c \cap B^c$

11) $A \subseteq B$ if and only if $B^c \subseteq A^c$.

Proof: Exercise! Very good practice

(TFAE)

Theorem 3: Let A, B and C be subsets of a Universal set U. The following are equivalent:

1) $A \subseteq B$
2) $A \cap B^c = \emptyset$
3) $A^c \cup B = U$

means (1) ↔ (2), (2) ↔ (3), (3) ↔ (1)

or also (1) ↔ (2) ↔ (3) ↔ (3) chain of logical equivalences

Technique: To prove that a sequence of statements are equivalent to each other, prove a sequence of conditionals so there is an unbroken chain of conditionals from each statement to any other statement.

E.g. To prove P, Q, R, S are equivalent (which means $P \leftrightarrow Q$, $P \leftrightarrow R$, $P \leftrightarrow S$, $Q \leftrightarrow R$, $Q \leftrightarrow S$, $P \leftrightarrow S$)

it is enough to show $P \to Q$, $Q \to R$, $R \to S$, $S \to P$.

also enough: $P \to Q$, $Q \to R$, $R \to P$, $S \leftrightarrow Q$

Proof of Thm 3: We will prove (1) $\longrightarrow$ (2) $\longrightarrow$ (3) $\longrightarrow$ (1)

[(1) $\to$ (2)] We proved in Thm 18.6 that if $A \subseteq B$ then $A \cap B^c = \emptyset$.

[(2) $\to$ (3)] Suppose $A \cap B^c = \emptyset$. We want to show $A^c \cup B = U$. But

$$A^c \cup B \overset{\text{AST (7)}}{=} A^c \cup (B^c)^c \overset{\text{De Morgan}}{=} (A \cap B^c)^c = \emptyset^c \overset{\text{AST (9)}}{=} U$$

So $A^c \cup B = U$, as desired

[(3) $\to$ (1)] Suppose $A^c \cup B = U$. We want to show $A \subseteq B$. Suppose $x \in A$. Then $x \notin A^c$. But certainly $x \in U = A^c \cup B$. Therefore x must be in B. It follows that $A \subseteq B$. $\square$

Cartesian Products (5.4)

Definition 4: Let A and B be sets. An **ordered pair** is a single pair of objects (a, b) with $a \in A$ and $b \in B$, where order matters.

- Equality for order pairs: $(a, b) = (a', b')$ if and only if $a = a'$ and $b = b'$.
- Coordinates: in the ordered pair (a, b), a is called the first coordinate. b is called the second coordinate